

Lecture 3, Mar 20 - 2026.

Arc Length.

We know that

$$x = r \cos \theta = f(\theta) \cos \theta$$

$$y = r \sin \theta = f(\theta) \sin \theta.$$

because $r = f(\theta)$. Then,

$$\frac{dx}{d\theta} = f'(\theta) \cos \theta - f(\theta) \sin \theta.$$

$$\frac{dy}{d\theta} = f'(\theta) \sin \theta + f(\theta) \cos \theta.$$

$$\begin{aligned} \text{So, } \left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 &= (f'(\theta) \cos \theta - f(\theta) \sin \theta)^2 \\ &\quad + (f'(\theta) \sin \theta + f(\theta) \cos \theta)^2 \\ &= [f'(\theta)]^2 \cos^2 \theta - 2f'(\theta)f(\theta) \cos \theta \sin \theta + [f(\theta)]^2 \sin^2 \theta \\ &\quad + [f'(\theta)]^2 \sin^2 \theta + 2f'(\theta)f(\theta) \cos \theta \sin \theta + [f(\theta)]^2 \cos^2 \theta \\ &= [f'(\theta)]^2 \{ \underbrace{\cos^2 \theta + \sin^2 \theta}_1 \} + [f(\theta)]^2 \{ \underbrace{\sin^2 \theta + \cos^2 \theta}_1 \} \\ &= [f'(\theta)]^2 + [f(\theta)]^2 \quad \text{since } r = f(\theta) \\ &= \left(\frac{dr}{d\theta}\right)^2 + r^2. \end{aligned}$$

$$L = \int_{\theta_0}^{\theta_f} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta.$$

$$L = \int_{\theta_0}^{\theta_f} \sqrt{\left(\frac{dr}{d\theta}\right)^2 + r^2} d\theta. \quad \text{for } \theta \in [\theta_0, \theta_f].$$

Example. Find the length of the cardioid. $r = 1 + \sin \theta$.

Sln. $L = \int_{\theta_0}^{\theta_f} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$. In this particular case $\theta \in [0, 2\pi]$.

$$\frac{dr}{d\theta} = \frac{d}{d\theta}(1 + \sin\theta) = \cos\theta. \quad \text{So}$$

$$\begin{aligned} L &= \int_0^{2\pi} \sqrt{(1 + \sin\theta)^2 + \cos^2\theta} \, d\theta = \int_0^{2\pi} \sqrt{1 + 2\sin\theta + \underbrace{\sin^2\theta + \cos^2\theta}_1} \, d\theta \\ &= \int_0^{2\pi} \sqrt{2 + 2\sin\theta} \, d\theta = \sqrt{2} \int_0^{2\pi} \sqrt{1 + \sin\theta} \, d\theta. \end{aligned}$$

Tangents. We know.

$$x = r \cos\theta = f(\theta) \cos\theta$$

$$y = r \sin\theta = f(\theta) \sin\theta$$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$$\frac{dx}{d\theta} = f'(\theta) \cos\theta - f(\theta) \sin\theta$$

$$\frac{dy}{d\theta} = f'(\theta) \sin\theta + f(\theta) \cos\theta$$

$$\frac{dy}{dx} = \frac{f'(\theta) \sin\theta + f(\theta) \cos\theta}{f'(\theta) \cos\theta - f(\theta) \sin\theta}$$

Example. a). For the cardioid $r = 1 + \sin\theta$ find the slope of the tangent line when $\theta = \pi/3$.

b). Find the points on the cardioid where the tangent line is horizontal or vertical.

Sln. a).

$$\frac{dy}{dx} = \frac{f'(\theta) \sin\theta + f(\theta) \cos\theta}{f'(\theta) \cos\theta - f(\theta) \sin\theta} \quad \text{but } f'(\theta) = \cos\theta$$

$$\frac{dy}{dx} = \frac{\cos\theta \sin\theta + (1 + \sin\theta) \cos\theta}{\cos^2\theta - (1 + \sin\theta) \sin\theta}$$

$$\frac{dy}{dx} = \frac{\cos\theta (1 + 2\sin\theta)}{1 - \sin^2\theta - \sin\theta - \sin^2\theta} = \frac{\cos\theta (1 + 2\sin\theta)}{1 - 2\sin^2\theta - \sin\theta}$$

$$m = \left. \frac{dy}{dx} \right|_{\theta = \pi/3} = \frac{\cos \pi/3 (1 + 2\sin \pi/3)}{1 - 2\sin^2 \pi/3 - \sin \pi/3} = \frac{\frac{1}{2} (1 + \sqrt{3})}{1 - \frac{3}{2} - \frac{\sqrt{3}}{2}}$$

$$m = \frac{1 + \sqrt{3}}{-1 - \sqrt{3}} = -1.$$

For you! try find the equation of the tangent line.

b). $\frac{dy}{d\theta} = \cos\theta(1+2\sin\theta) = 0$ for $\theta \in [0, 2\pi]$.

implies that $\cos\theta = 0$ or $\sin\theta = -\frac{1}{2}$.

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$; r = 1 + \sin\theta.$$

if $\theta = \frac{\pi}{2}$ then $r = 2$. $(2, \frac{\pi}{2})$.

if $\theta = \frac{3\pi}{2}$ then $r = 0$. $(0, \frac{3\pi}{2})$.

if $\theta = \frac{7\pi}{6}$ then $r = \frac{1}{2}$. $(\frac{1}{2}, \frac{7\pi}{6})$.

if $\theta = \frac{11\pi}{6}$ then $r = \frac{1}{2}$. $(\frac{1}{2}, \frac{11\pi}{6})$.

$$\begin{aligned} \frac{dx}{d\theta} &= 1 - 2\sin^2\theta - \sin\theta = -(2\sin^2\theta + \sin\theta - 1) \\ &= -\frac{(4\sin^2\theta + 2\sin\theta - 2)}{2} \\ &= -\frac{(2\sin\theta + 2)(2\sin\theta - 1)}{2} \\ &= -(\sin\theta + 1)(2\sin\theta - 1) \\ &= (1 - 2\sin\theta)(1 + \sin\theta) = 0 \end{aligned}$$

$$\sin\theta = \frac{1}{2} \quad \text{or} \quad \sin\theta = -1.$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6} \quad \theta = \frac{3\pi}{2}.$$

$$\lim_{\theta \rightarrow \frac{3\pi}{2}} \frac{dy}{dx} = \left(\lim_{\theta \rightarrow \frac{3\pi}{2}} \frac{1 + 2\sin\theta}{1 - 2\sin\theta} \right) \cdot \left(\lim_{\theta \rightarrow \frac{3\pi}{2}} \frac{\cos\theta}{1 + \sin\theta} \right)$$

$$L'H = -\frac{1}{3} \lim_{\theta \rightarrow \frac{3\pi}{2}} \frac{-\sin\theta}{\cos\theta} = -\infty.$$

at $\theta = \frac{3\pi}{2}$ we have a tangent vertical line.

For you! Check for $\theta = \frac{\pi}{6}, 5\frac{\pi}{6}$.